1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In HR+/HER2− breast cancers with high tumor-infiltrating lymphocytes (TIL-high), concurrent dominance of MARCO-expressing tumor-associated macrophages (TAMs) and a tumor transcriptomic TP53-mutant-like program define a functionally immunosuppressive microenvironment that blunts the therapeutic benefit of endocrine therapy (ET). Specifically: (i) this triad (TIL-high + MARCO-high TAMs + TP53-mutant-like signature) will correlate with higher ET resistance rates and shorter time-to-progression compared with TIL-high tumors lacking MARCO dominance or TP53-mutant-like transcriptional features; (ii) mechanistically, MARCO+ TAMs suppress antigen presentation (lower MHC-I/MHC-II and co-stimulatory signals), secrete immunosuppressive cytokines, and interact with TP53-contextual programs to dampen interferon signaling and T cell–mediated anti-tumor activity; (iii) disrupting MARCO+ TAMs or reprogramming TP53-associated immunomodulation will restore antigen presentation and T cell function, thereby improving ET response in preclinical models and in prospective translational studies. Therefore, a minimal core readout set—imaging-based TIL density plus MARCO expression and a TP53-mutant-like transcriptional score—will identify the patient subset most likely to exhibit ET resistance and who may benefit from macrophage-targeted or TP53-contextualized combination therapies to restore endocrine sensitivity.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MARCO, RNA-based TP53 mutant-like
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MARCO: MARCO) do not uniformly translate into better outcomes under endocrine therapy; what governs when immune infiltration is functionally beneficial versus neutral or deleterious? 2) Tier 1 basis: I built on the MARCO path (immune/macrophage axis linked to endocrine/metabolic therapy context) and the RNA-based TP53 mutant-like path (transcriptomic TP53 dysfunction that can reprogram tumor-intrinsic immunomodulatory programs), because together they offer a mechanistic route linking macrophage-mediated suppression and tumor-intrinsic immune evasion.

RNA-based TP53 mutant-like: RNA-based TP53 mutant-like path (transcriptomic TP53 dysfunction that can reprogram tumor-intrinsic immunomodulatory programs), because together they offer a mechanistic route linking macrophage-mediated suppression and tumor-intrinsic immune evasion.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	HIF1A	Supporting modifier from knowledge graph
Secondary	CD4 (21-gene immune signal)	Supporting modifier from knowledge graph
Secondary	estrone	Supporting modifier from knowledge graph
Secondary	testosterone	Supporting modifier from knowledge graph
Secondary	PepQuant_nine_biomarkers	Supporting modifier from knowledge graph
Secondary	Recurrence_Score (RS)	Supporting modifier from knowledge graph
Secondary	FACT-G (QoL / adherence proxy)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: The interaction between the imaging feature and anchor gene predicts clinical outcome.

Prediction 2: Spatial analysis will confirm the mechanistic link at the tissue level.

Prediction 3: Functional perturbation will establish causality.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- HIF1A: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CD4 (21-gene immune signal): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.